CEBU INSTITUTE OF TECHNOLOGY - UNIVERSITY
College of Computer Studies, IT411 Capstone & Research 2

Final Validation Evidence, Results & Recommendations Templates

Week 12 Deliverables 03, 04 & 05: testing session logs, results report, and improvements

Course & Section:	IT411 (Section G7) — Capstone & Research 2 | Semester 1, AY 2026–2027
Group Code / Team:	`2627-sem1-it411-56` — Group 56 (`Riva, Zendrix`; `Palis, John James`; `Miel, Kaysean`; `Durano, Ashlee Skye`; `Bien, Erik Samuel`)
Adviser:	Mr. Joemarie C. Amparo | **Course Instructor:** Mr. Ralph P. Laviste
Deliverable Reference:	Week 12 Deliverables 3, 4 & 5 (`03_Testing_Evidence`, `04_Results_and_Analysis`, `05_Recommendations_and_Improvements`, Due October 31, 2026)

Folder 03 Template: `03_Testing_Evidence/TESTING_EVIDENCE_INDEX.md`

1. Summary of Executed Validation Sessions

- **Software Version Evaluated:** `playIT v1.2.0-rc2` (`refactor/hear-say-it`, offline production APK)
- **Test Device(s):** Samsung Galaxy A21s (`SM-A217F`, Android 12, 3 GB RAM, Airplane Mode ON)
- **Data Collection Period:** October [__] – October [__], 2026 (Post-Adviser Verification Approval)

2. Evidence Inventory & Anonymized Session Log

Evidence Folder / File	Description	Respondent Codes Covered	Linked Instrument
`03_Testing_Evidence/Teacher_Audits/`	Completed Gate 3 Blind 26-Phoneme & 33-Word CVC Audit sheets	`T-01` .. `T-05`	`Instrument A`
`03_Testing_Evidence/Child_Observation_Logs/`	Completed Silent-Observer Navigation Sheets & Blind Dual-Teacher `Say It` Rating Logs	`C-01` .. `C-20`	`Instrument B`
`03_Testing_Evidence/System_Telemetry_CSVs/`	Exported Room SQLite CSV dumps (`say_it_attempts.csv`, `find_it_attempts.csv`, `blend_it_attempts.csv`, `lesson_progress.csv`) & generated A4 PDF Progress Reports	`C-01` .. `C-20`	`Instrument C`
`03_Testing_Evidence/Hardware_QA_Logs/`	ADB latency logs, `gfxinfo` 60 FPS dumps, `meminfo` dumps, and 20× force-close persistence log on `SM-A217F`	`QA-01` .. `QA-03`	`Instrument D`
`03_Testing_Evidence/Stakeholder_Surveys/`	Child Smileyometer/Exit Choice sheets (`Part E1`) & Teacher/Parent SUS + TAM response records (`Part E2`)	`C-01`..`C-20`, `T-01`..`T-05`, `P-01`..`P-10`	`Instrument E`

Folder 04 Template: `04_Results_and_Analysis/FINAL_VALIDATION_RESULTS_AND_ANALYSIS.md`

1. Respondent Profile & Sample Distribution

Cohort	Target Role	Actual N	Demographic / Qualification Summary
Cohort 1 (`C-01`..`C-__`)	Grade 1 Learners (Primary End-Users)	[__]	Ages 6–7 ([__] age 6, [__] age 7); Grade 1 emergent readers.
Cohort 2 (`T-01`..`T-__`)	DepEd Grade 1 Reading Teachers	[__]	Mean teaching experience: [__] years in Grade 1 Reading / MTB-MLE / English.
Cohort 3 (`P-01`..`P-__`)	Parents / Primary Caregivers	[__]	Parents/guardians of Grade 1 learners supervising home reading practice.
Cohort 4 (`QA-01`..`QA-03`)	Technical / QA Evaluators	3	Hardware & accessibility benchmark evaluators on `SM-A217F`.

2. Empirical Results by Research Question & SMART Goal

2.1 RQ1 — Pedagogical & Acoustic Integrity (SMART-01, SMART-02, HI-1, HI-2)

- **Gate 3 Blind Phoneme Audit (Instrument A Part A1):** [__] / 26 ([___]%) phoneme clips transcribed accurately and rated schwa-free (>= 2/3 teachers).
- **Decodable CVC Word Bank Audit (Instrument A Part A2):** [__] / 33 ([___]%) Blend It words verified as 100% cumulatively decodable.
- **Child Hear It Independent Completion (Instrument B Part B1):** [__] / [__] sessions ([___]%, 95% Wilson CI [___%, ___%]) completed Hear It with 0 adult prompts *(Target: ≥ 90.0%).*

2.2 RQ2 — Offline Speech Recognition & Articulatory Scaffolding (SMART-03, SI-1–SI-4)

Metric	Formula	Empirical Result	95% Wilson CI / Spread	Target Criterion	Verdict
Teacher-ASR Agreement (P_(agree))	(TP + TN) / N_(trials)	[___]% ([__]/[__])	[___%, ___%]	**≥ 80.0%**	[PASS / FAIL]
False-Reject Rate (P_(FR))	FN / (TP + FN)	[___]% ([__]/[__])	[___%, ___%]	**≤ 15.0%**	[PASS / FAIL]
Dual-Teacher Inter-Rater kappa	Cohen's kappa	[0.__]	—	**≥ 0.70**	[PASS / FAIL]
3-Attempt Lesson Cleared Rate (SI-1)	Cleared ≤ 3 att. / Total	[___]%	[___%, ___%]	**≥ 80.0%**	[PASS / FAIL]
Attempt 2/3 Mouth-Cue Recovery (SI-2)	Att. 2–3 pass mid Att. 1 miss	[___]%	[___%, ___%]	Documented > 0%	[PASS / FAIL]
Say It **P90 Latency (D-01)**	P90(t_(feedback) − t_(EOS))	[___] ms	P50: [___] ms	**≤ 500 ms**	[PASS / FAIL]

2.3 RQ3 — Child Usability, Discrimination & Blending Mastery (SMART-04, SMART-05, 03)

- **Zero-Prompt Child Screen Navigation (03):** [___]% ([__]/[__] transitions) completed without adult intervention *(Target: ≥ 85.0%).*
- Find It **First-Tap Accuracy & Latency (SMART-04, D-02):** First-tap accuracy [___]%; P90 tap latency [___] ms *(Target: ≤ 300 ms).*
- Blend It **CVC Word Completion (SMART-04):** [___]% of learners solved all 3 group CVC words with ≥ 2star mastery.
- **Child Affect & Voluntary Replay (Instrument E Part E1):** [___]% selected Happy Smile (3/3); [___]% voluntarily chose to play another letter when offered the exit prompt.

2.4 RQ4 — Hardware Performance, Offline Persistence & Privacy (SMART-06, 04, 05)

- **20-Cycle Force-Close Persistence (D-05):** [20 / 20] (100%) cycles retained exact SQLite state in Airplane Mode.
- **Airplane-Mode Module Operability (D-06):** [4 / 4] learning modules + offline PDF export (100%) operated without network connectivity.
- **Frame Stability & Memory (D-04):** Cold start [___] ms (≤ 2500 ms); Jank [___]% (≤ 10%); Peak PSS RAM [___] MB (≤ 200 MB).

2.5 RQ5 — Stakeholder Usability (SUS), TAM & Diagnostic Utility (SMART-07, 06)

Stakeholder Scale / Dimension	Teachers (n = [__]) Mean (SD)	Parents (n = [__]) Mean (SD)	Combined (N = [__]) Mean (SD)	Target Criterion	Verdict
Brooke System Usability Scale (SUS, 0–100)	[___._] ([_._])	[___._] ([_._])	[___._] ([_._])	**≥ 75.0**	[PASS / FAIL]
TAM Perceived Usefulness (PU1–PU4, 1–5)	[_. __] ([_.__])	[_. __] ([_.__])	[_. __] ([_.__])	**≥ 4.00 / 5.00**	[PASS / FAIL]
TAM Perceived Ease of Use (PEOU1–PEOU3, 1–5)	[_. __] ([_.__])	[_. __] ([_.__])	[_. __] ([_.__])	**≥ 4.00 / 5.00**	[PASS / FAIL]
Pedagogical Alignment (PA1–PA5, 1–5)	[_. __] ([_.__])	N/A	[_. __] ([_.__])	**≥ 4.00 / 5.00**	[PASS / FAIL]
Adult Diagnostic & PDF Task Success (Tasks A1–A4)	100%	100%	100%	**100%**	[PASS / FAIL]

3. Comparison: Initial MVP Validation (Weeks 1–2) vs. Final System Validation (Week 12)

Evaluation Dimension	Initial MVP Baseline (Weeks 1–2, N = 18)	Final Refactored System (Week 12, v1.2.0-rc2)	Net Improvement & Engineering Root Cause
Phoneme Audio Purity (Gate 3)	3.10 / 5.00 (Schwa /■/ on stop consonants)	[___]% **(26/26 Pass)**	Audited human/Kokoro/Chatterbox/ElevenLabs clips (Cards 09a–09c, 15b, 19).
Say It **ASR Child Agreement**	2.85 / 5.00 (High false-rejects & foil acceptance)	[___]% **Agreement /** [___]% **FRR**	Constrained SpeechValidator grammar + foil rejection + non-punitive 3-attempt scaffolding (Cards 02, 05, 13, 21).
Word Blending (Blend It**)**	Absent in MVP (0 words)	**33 Decodable CVC/VVC Words across 6 Groups**	Full BlendItScreen + non-solved tile deal guard (Cards 08, 15, 25, 31f).
Child Touch & Stage Ergonomics	3.20 / 5.00 (Sub-48dp buttons, competing gradients)	**100% ≥ 64 dp targets + Duolingo ABC Grounded Stages**	LessonScaffold, 64dp child controls, Argon 4 illustrations, and Card 32 grounded horizon stages.

Folder 05 Template: 05_Recommendations_and_Improvements/RECOMMENDATIONS_AND_POST_VALIDATION_IMPROVEMENTS.md

1. Issues & Concerns Raised by Users or Stakeholders During Final Validation

Issue ID	Source Cohort (C, T, P, QA)	Module / Screen Affected	Description of Observed Issue or Stakeholder Feedback	Severity (Critical, Major, Minor, Enhancement)
FV-01	[____]	[____]	[Populate from Instrument B / Instrument E OE-02 findings]	[____]
FV-02	[____]	[____]	[Populate from Instrument B / Instrument E OE-02 findings]	[____]
FV-03	[____]	[____]	[Populate from Instrument B / Instrument E OE-02 findings]	[____]

2. Prioritized Recommended Improvements & Corrective Actions Taken

Priority Rank	Issue ID	Recommended Improvement / Corrective Action	Action Status (Implemented in Code, Documented in SRS/SDD, Scheduled for Future Work)	Verification / Commit Reference
P1 (High)	FV-01	[Action description]	[Status]	[Commit / Test ID]
P2 (Medium)	FV-02	[Action description]	[Status]	[Commit / Test ID]
P3 (Low)	FV-03	[Action description]	[Status]	[Commit / Test ID]

3. Updates Required to Software, Technical Documentation, or Test Cases

- **Source Code Updates:** [List any post-validation hotfixes or tuning commits]
- **SRS / SDD / SPMP / STD / RTM Updates:** [List section revisions reflecting final validation outcomes]

4. Remaining Scope Limitations & Recommendations for Future Research

1. **Dialectal & Multilingual Expansion (Cebuano / Bisaya & Tagalog MTB-MLE):** While playIT v1.2.0 targets the 26 English alphabet phonemes and 33 English CVC words using a Marungko-inspired sequence, future iterations can add dedicated Bisaya and Filipino orthography modules.
2. **Longitudinal Classroom Retention Study:** Extending the 7-day retention window (RetentionCalculator) to a full 8-week DepEd grading quarter across multiple public schools to measure longitudinal reading fluency gains (EGRA / Phil-IRI alignment).
3. **Adaptive Acoustic Calibration for High-Noise Public Classrooms:** Expanding the ambient noise detector in SayItViewModel with adaptive SNR thresholds for 40-student classrooms without external lapel microphones.